

Luca Nepote *PhD Researcher | Machine Learning*

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PhD researcher in Machine Learning, working on generative models and trajectory inference. Background spanning data science, quantum computing, and physics. Seeking a 3–6 month research internship.

Professional Research Experience

PhD Researcher, *Eurecom (ProbAI Lab)* 📍 – Sophia Antipolis, France Jun 2025 – Ongoing

- Working on Generative Models and Trajectory Inference methods for biological applications.
- Developed a novel methodology to evaluate trajectory inference methods in function space. 📄

Quantum AI Research Intern, *Amadeus IT (Research Team)* 📍 – Villeneuve Loubet, France Sep 2024 – Feb 2025

- Explored Variational Quantum Algorithms for Quantum Machine Learning to enhance AI energy scalability.
- Improved accuracy by 10% integrating Variational Quantum Algorithm and Variational Inference with Binary Neural Networks. 📄

Research Intern, *INRiM (National Institute of Metrological Research)* 📍 – Turin, Italy Nov 2021 – Jun 2022

- Explored and automated mathematical models for magnetic hyperthermia, simulating pre-clinical tests for different nanoparticles and enhancing research efficiency.
- Used ML models in Matlab to generalize results to unseen data, reducing the need for additional real-world tests.

Publications

- C. Wang*, **L. Nepote***, G. Franzese, P. Michiardi. *Relative Entropy Estimation in Function Space: Theory and Applications to Trajectory Inference*. ICML 2026 (Main Conference), Seoul. 📄 *equal contribution
- **L. Nepote**, A. Lhéritier, N. Bondoux, M. Kountouris, M. Filippone. *Variational Inference for Quantum HyperNetworks*. IJCNN 2025, Quantum Workshop (Oral), Rome. 📄

Education

PhD, Machine Learning, *Sorbonne Université* – Sophia Antipolis, France 2025 – Ongoing

- Hosted at Eurecom, ProbAI Lab (Alpha AI Center) 📍. Advisors: P. Michiardi 📄, G. Franzese 📄.

MSc, Data Science, *Eurecom* – Sophia Antipolis, France 2023 – 2025

- Double Degree in Data Science and Telecommunications Engineering.

MSc, Telecommunications Engineering, *Polytechnic of Turin* – Turin, Italy 2022 – 2025

- Grade: 110/110 *cum laude*. Double Degree in Data Science and Telecommunications Engineering.

BSc, Physics Engineering, *Polytechnic of Turin* – Turin, Italy 2019 – 2022

- Grade: 108/110. Thesis: “Numerical Modeling of Magnetic Resonance Imaging: Calculation of T2 Parameter of Tissues”, INRiM.

Skills

Programming & Core: Python, Matlab

ML & Frameworks: PyTorch, TensorFlow, Jax, Qujax, Qiskit

Model Architectures: Generative/Diffusion Models, VAEs, Variational Inference, HyperNetworks

Tools: Git/GitHub, Linux/Unix, LaTeX

Languages: Italian (Native), English (Professional), French (Intermediate)

Courses & Certifications

- Qiskit Global Summer School, 2024, “Path to Utility” — IBM
- Qiskit Global Summer School, 2023, “From Theory to Implementation” — IBM
- CODAS Summer School, 2023, BIP ERASMUS+ - Aalto University
- Project Management, 2024 — Eurecom

Scholarships

- Scholarship, Double Degree Program (Polytechnic of Turin – Eurecom).
- Scholarship, CODAS Summer School, Aalto University.
- Tuition Reduction for Merit, Polytechnic of Turin.